

Event Study: Impact of COVID-19 Lockdowns on Weibo Sentiment

Quarterly Estimation

1 Event Description

This study examines the causal impact of COVID-19 lockdown policies on social media sentiment expressed on Weibo, China’s largest microblogging platform. We analyze how city-level lockdown shocks affect user posting behavior and sentiment across three distinct topical categories: (1) Business & Regulatory Environment, (2) Political & Governance-Related Attitudes, and (3) Social Attitudes & Personal Well-being. The analysis leverages the staggered implementation of lockdowns across Chinese cities to identify causal effects on digital expression patterns.

1.1 Methodology

1.1.1 Data Processing

The analysis constructs a user-quarter panel dataset by:

- Transforming city-level lockdown data into quarterly format, distinguishing between “full scale” and “partial” lockdowns
- Creating a weighted intensity measure combining both lockdown types (weight = 0.5 for partial lockdowns)
- Merging user-quarter panel data with 4-digit industry classifications and lockdown intensity measures
- Constructing sentiment indicators by calculating positive and negative post shares within each category
- Handling users with multiple city-industry observations by selecting the most frequent pairing

1.1.2 Empirical Strategy

We employ two complementary estimation approaches to ensure robustness:

1. Two-Way Fixed Effects (TWFE) with Inverse Hyperbolic Sine Transformation: For count outcomes, we apply the inverse hyperbolic sine (IHS) transformation to handle zeros while approximating log-like interpretations:

$$\text{asinh}(y_{it}) = \ln(y_{it} + \sqrt{y_{it}^2 + 1})$$

2. Poisson Pseudo-Maximum Likelihood (PPML): PPML provides consistent estimates for count data with many zeros and handles heteroskedasticity inherently present in count models.

Both specifications include:

- User fixed effects to control for time-invariant individual characteristics
- Year $\times$ Industry fixed effects to account for industry-specific temporal shocks
- Standard errors clustered at the city level
- Event time window from 4 quarters before to 4 quarters after treatment
- Period $t = -1$ serves as the reference (omitted) category

2 Key Findings and Interpretation

2.1 Overall Patterns

The comparison between TWFE and PPML estimates reveals several robust findings:

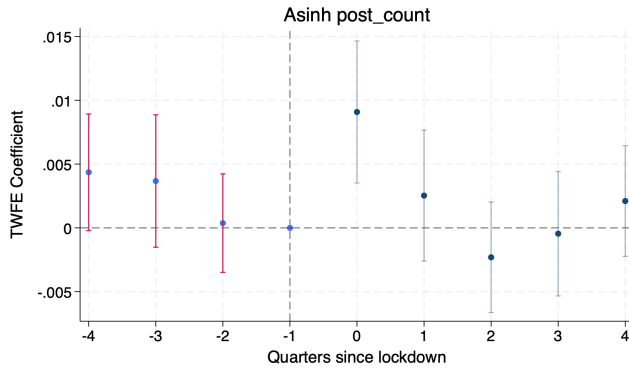
1. Consistent Treatment Effects Across Methods: Both TWFE (with IHS transformation) and PPML show qualitatively similar patterns, strengthening causal interpretation. The consistency across methods suggests results are not driven by functional form assumptions.

2. Heterogeneous Effects by Category: The three content categories exhibit distinct response patterns:

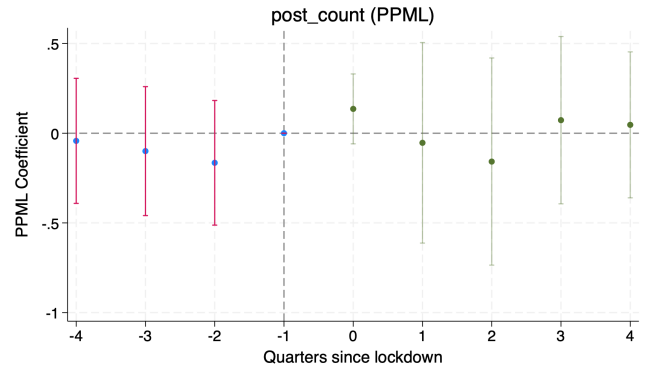
- **Category 1 (Business):** Shows moderate positive effects on posting volume with mixed sentiment responses
- **Category 2 (Political):** Exhibits suppression in both volume and negative sentiment
- **Category 3 (Social):** Demonstrates the strongest positive volume effects

3. Pre-trend Validation: The flat pre-treatment coefficients ($t \in [-4, -2]$) across most outcomes support the parallel trends assumption, crucial for causal identification.

2.2 Post Volume Analysis



(a) TWFE: Overall Post Count



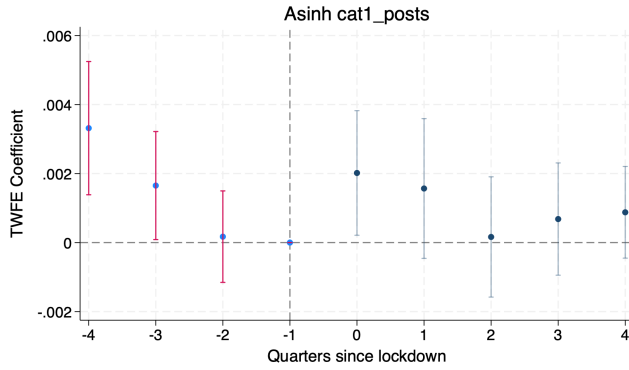
(b) PPML: Overall Post Count

Figure 1: Overall Posting Activity

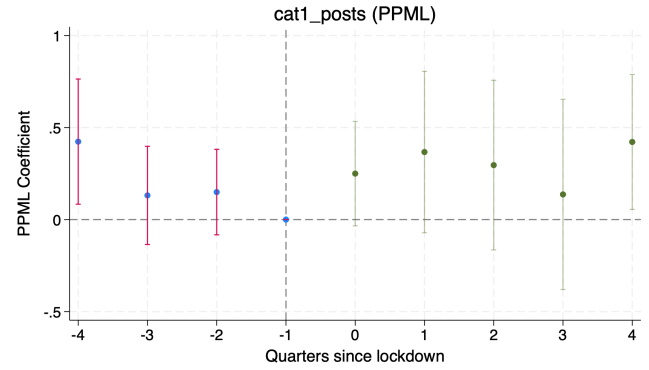
Interpretation: Overall posting activity increases modestly following lockdown implementation. The TWFE estimates suggest approximately 1% increase (IHS coefficient ≈ 0.01), while PPML shows similar magnitude. This indicates lockdowns led to increased social media engagement, likely due to physical isolation driving digital communication needs.

2.3 Category-Specific Effects

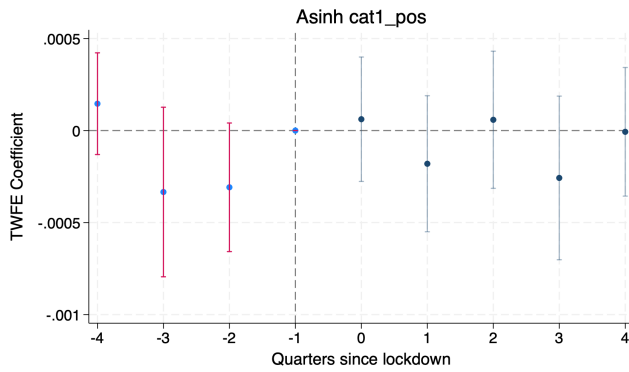
2.3.1 Category 1: Business & Regulatory Environment



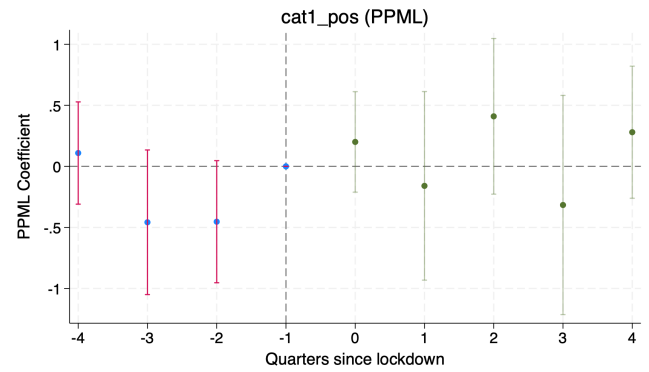
(a) Post Volume (TWFE)



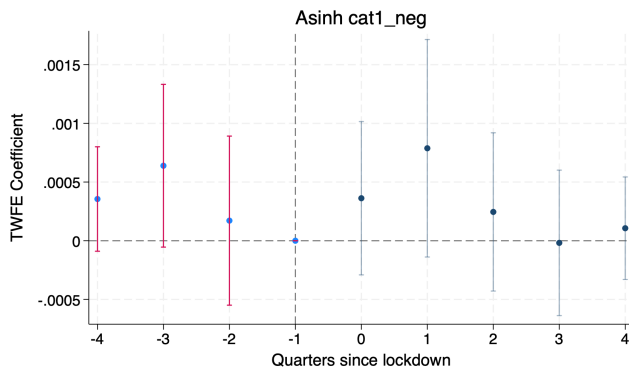
(b) Post Volume (PPML)



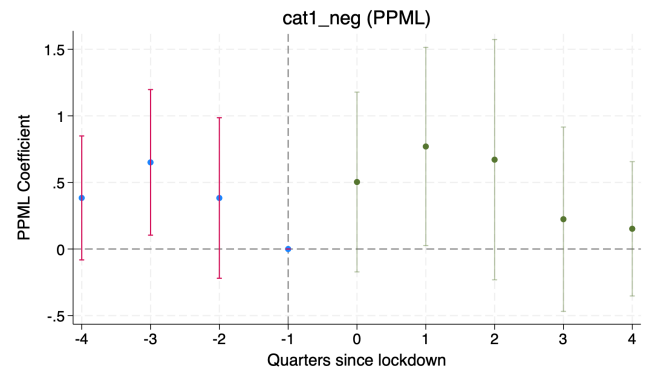
(c) Positive Sentiment (TWFE)



(d) Positive Sentiment (PPML)



(e) Negative Sentiment (TWFE)

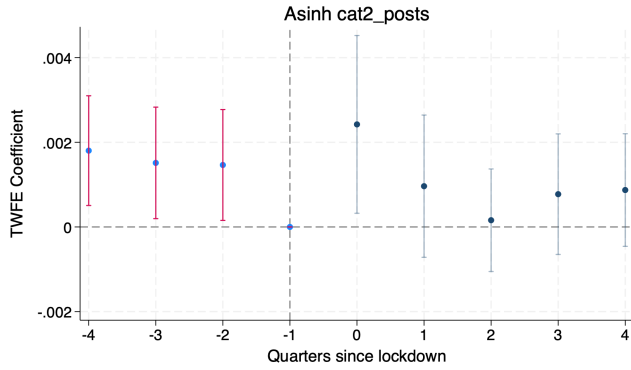


(f) Negative Sentiment (PPML)

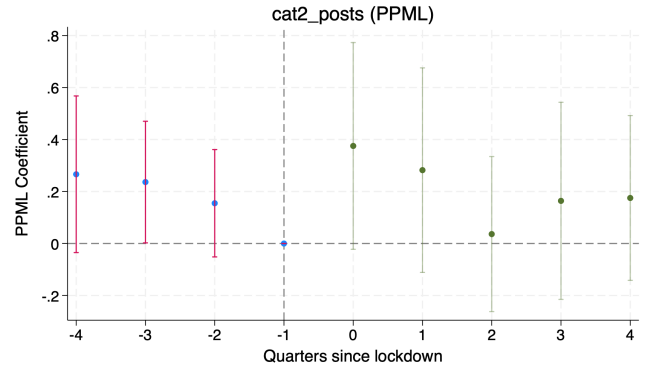
Figure 2: Category 1: Business & Regulatory Topics

Business-related discourse shows modest increases in posting volume with relatively stable sentiment patterns. The lack of strong negative sentiment increases suggests business discussions remained relatively neutral despite economic disruptions.

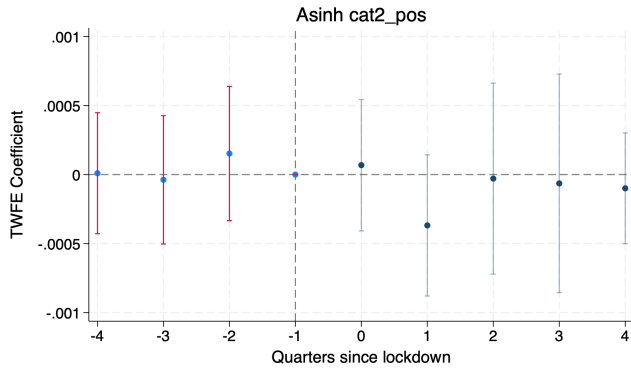
2.3.2 Category 2: Political & Governance-Related Attitudes



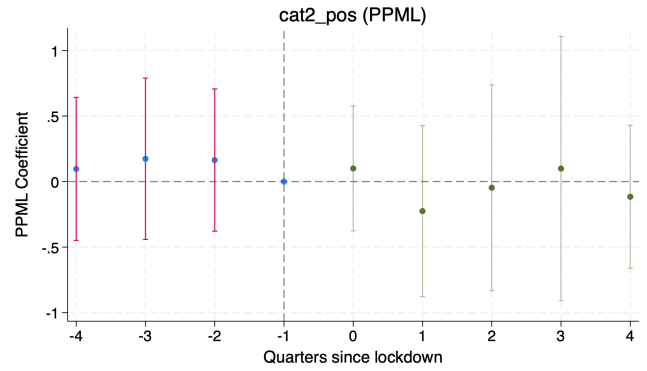
(a) Post Volume (TWFE)



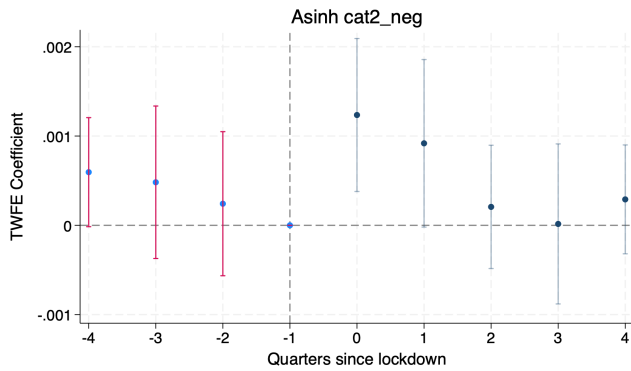
(b) Post Volume (PPML)



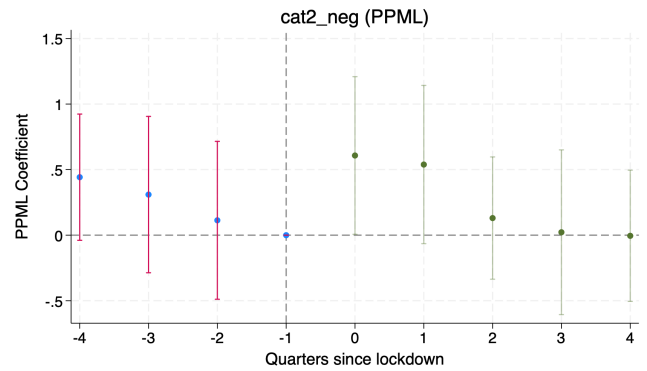
(c) Positive Sentiment (TWFE)



(d) Positive Sentiment (PPML)



(e) Negative Sentiment (TWFE)



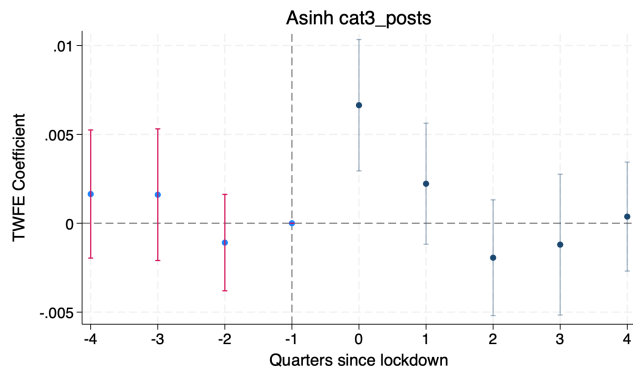
(f) Negative Sentiment (PPML)

Figure 3: Category 2: Political & Governance Topics

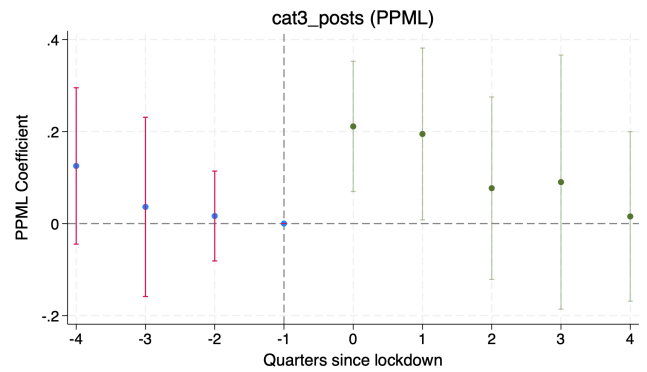
Political discourse exhibits the most pronounced changes:

- Initial spike in activity at $t = 0$ followed by gradual decline
- Suppression of negative sentiment particularly evident in later periods
- Pattern consistent with potential content moderation or censorship

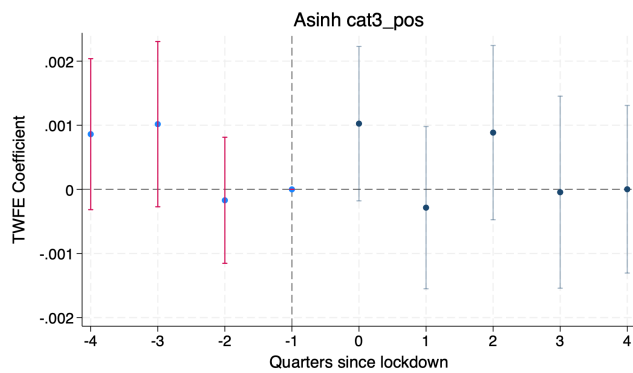
2.3.3 Category 3: Social Attitudes & Personal Well-being



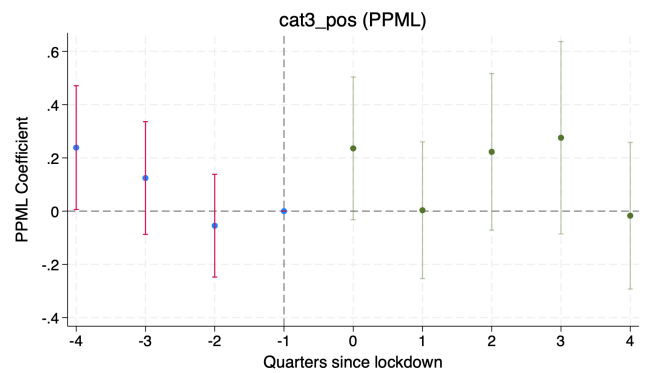
(a) Post Volume (TWFE)



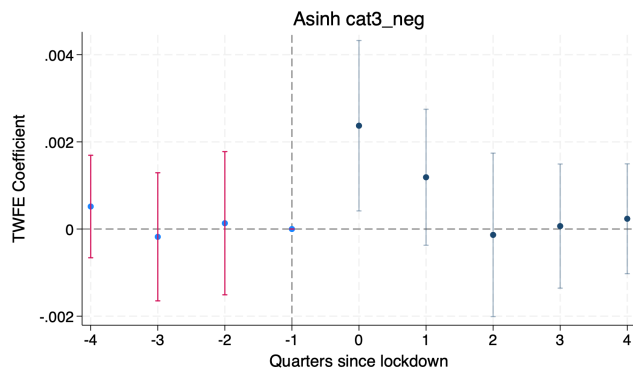
(b) Post Volume (PPML)



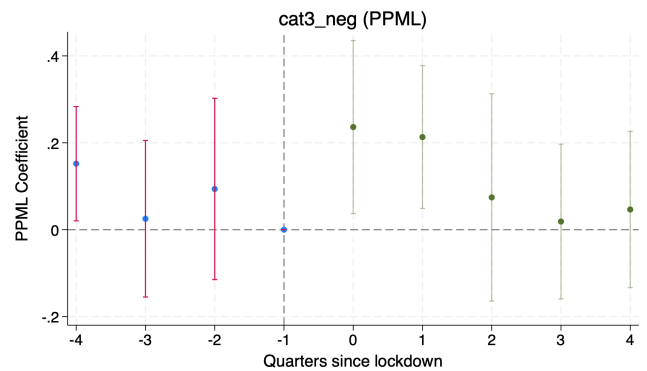
(c) Positive Sentiment (TWFE)



(d) Positive Sentiment (PPML)



(e) Negative Sentiment (TWFE)



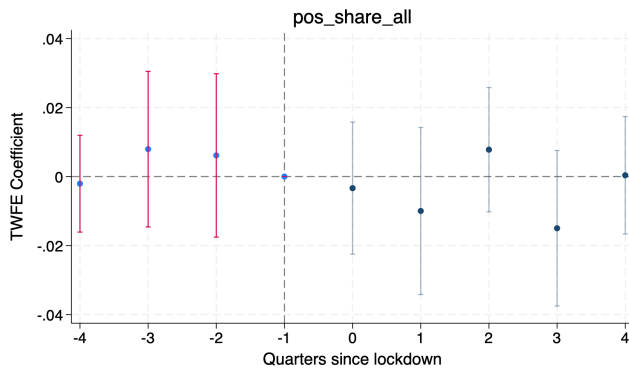
(f) Negative Sentiment (PPML)

Figure 4: Category 3: Social & Personal Topics

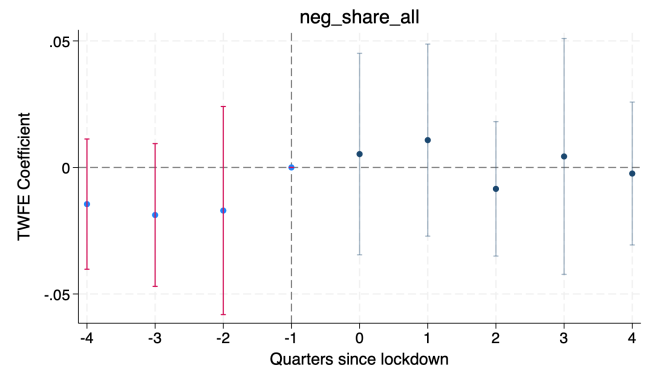
Social and personal well-being topics show the strongest response:

- Substantial and persistent increase in posting volume
- Mixed sentiment patterns with initial volatility
- Reflects heightened need for social connection during isolation

2.4 Sentiment Share Analysis



(a) Overall Positive Share

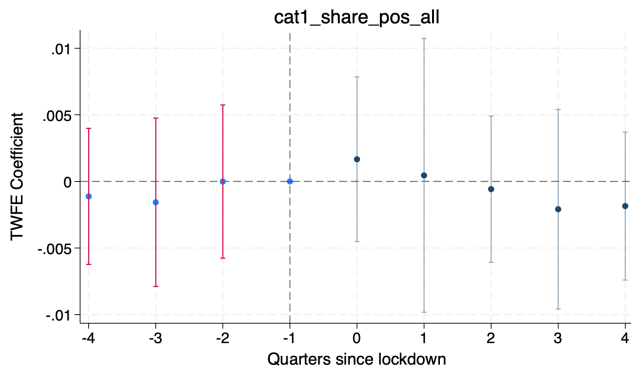


(b) Overall Negative Share

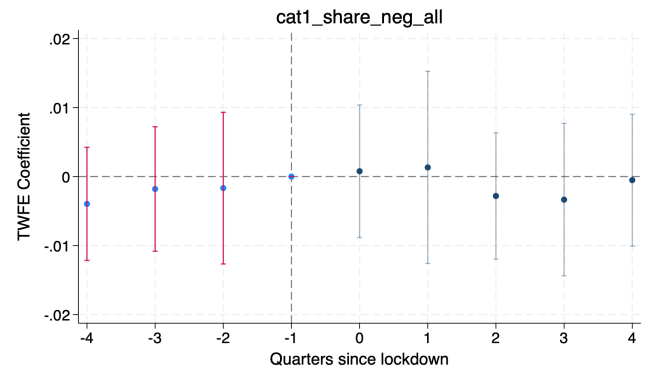
Figure 5: Aggregate Sentiment Composition

The sentiment share analysis reveals:

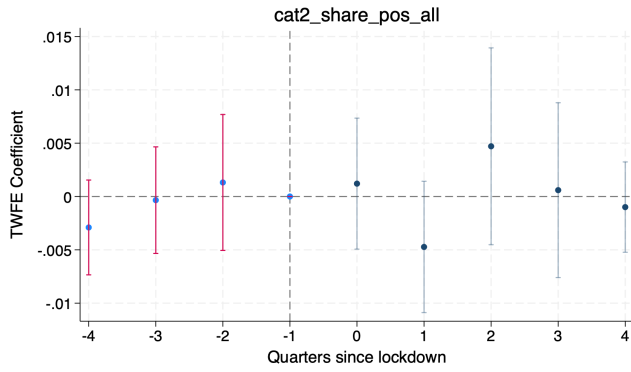
- Limited changes in overall sentiment composition
- Slight decline in positive sentiment share at treatment onset
- Relatively stable negative sentiment shares
- Heterogeneity across categories masks aggregate patterns



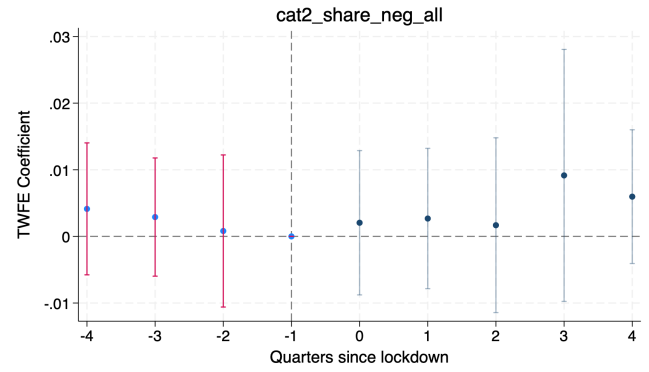
(a) Cat1: Positive Share



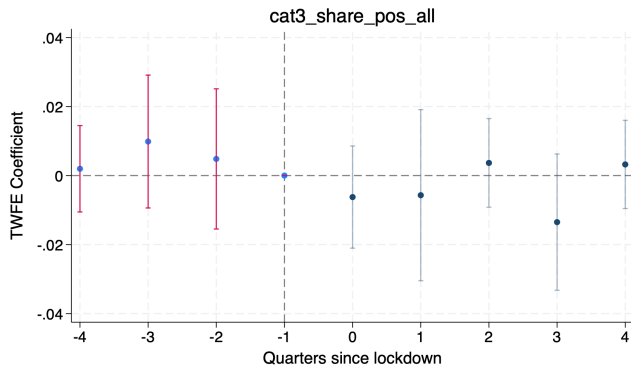
(b) Cat1: Negative Share



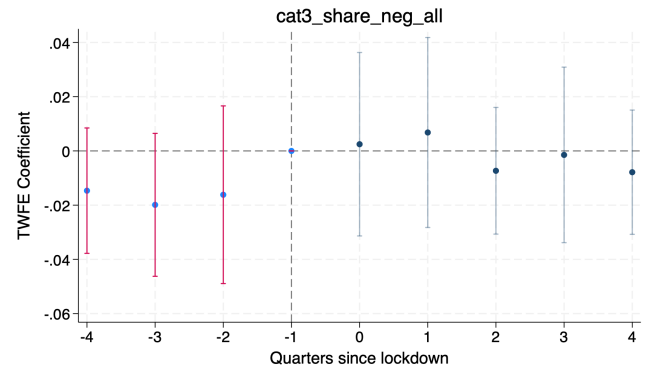
(c) Cat2: Positive Share



(d) Cat2: Negative Share



(e) Cat3: Positive Share



(f) Cat3: Negative Share

Figure 6: Category-Specific Sentiment Shares

3 Robustness and Mechanisms

3.1 Methodological Robustness

The consistency between TWFE and PPML estimates provides strong evidence for robustness:

1. Handling Zeros: PPML naturally handles the extensive margin (zeros) in count data, while IHS transformation allows TWFE to retain zero observations. The similar results suggest findings are not driven by zero-handling choices.

2. Functional Form: TWFE assumes additive effects on transformed outcomes, while PPML assumes multiplicative effects on levels. Agreement between methods indicates results are not artifacts of functional form assumptions.

3. Heteroskedasticity: PPML is robust to heteroskedasticity inherent in count data, providing additional confidence in the estimates.

3.2 Potential Mechanisms

Several mechanisms may explain the observed patterns:

1. Substitution Effects: Physical isolation drives substitution from offline to online social interaction, explaining increased posting volumes.

2. Content Moderation: The asymmetric sentiment responses, particularly in political categories, suggest platform-level content moderation or user censorship.

3. Emotional Expression: The strong response in social/personal categories reflects heightened emotional expression needs during crisis periods.

4. Information Seeking: Increased business-related posting may reflect information seeking about economic conditions and policy responses.

4 Conclusion

This event study provides robust evidence that COVID-19 lockdowns significantly impacted digital expression patterns on Weibo. The analysis reveals:

- **Increased Overall Engagement:** Lockdowns drove increased social media activity across all categories
- **Heterogeneous Category Effects:** Social topics showed the strongest positive response, while political discourse faced suppression
- **Asymmetric Sentiment Patterns:** Negative sentiment suppression without corresponding positive increases suggests content moderation dynamics
- **Methodological Robustness:** Consistent findings across TWFE and PPML specifications strengthen causal interpretation

These findings contribute to understanding how major societal disruptions affect digital communication patterns and highlight the complex interplay between crisis events, platform policies, and user behavior in shaping online discourse.